

Conceptual Models

Bonita Sharif

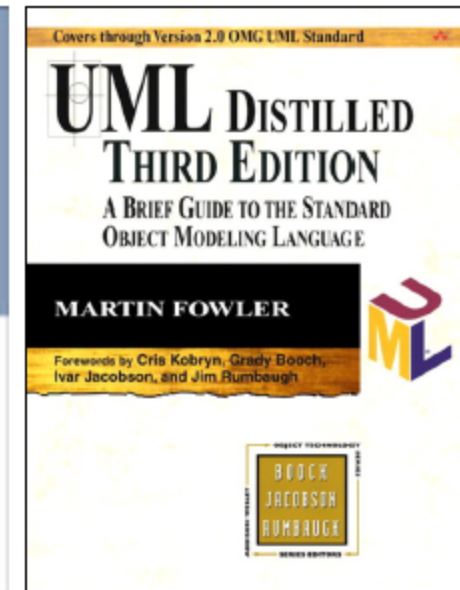
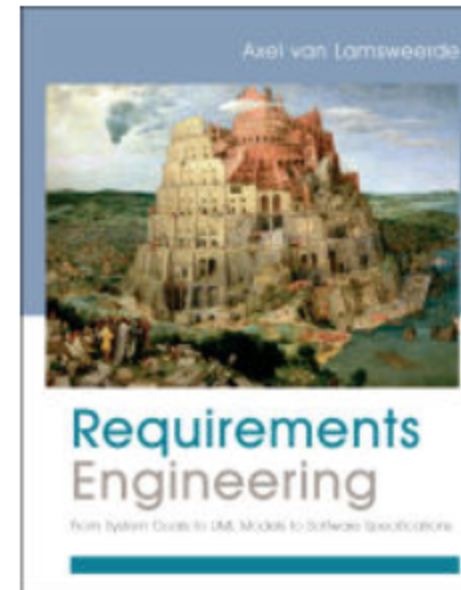
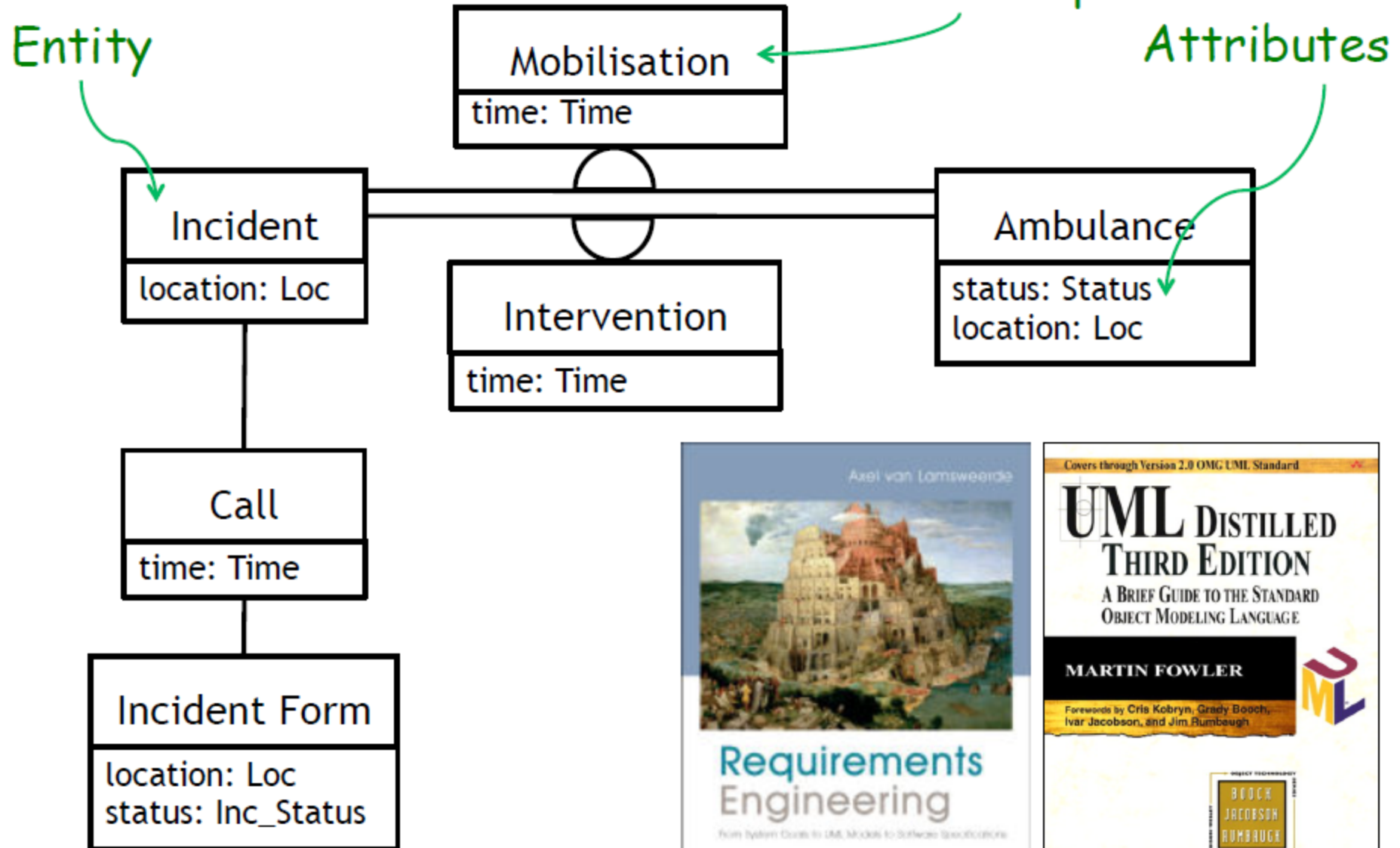
School of Computing

University of Nebraska - Lincoln

Motivation

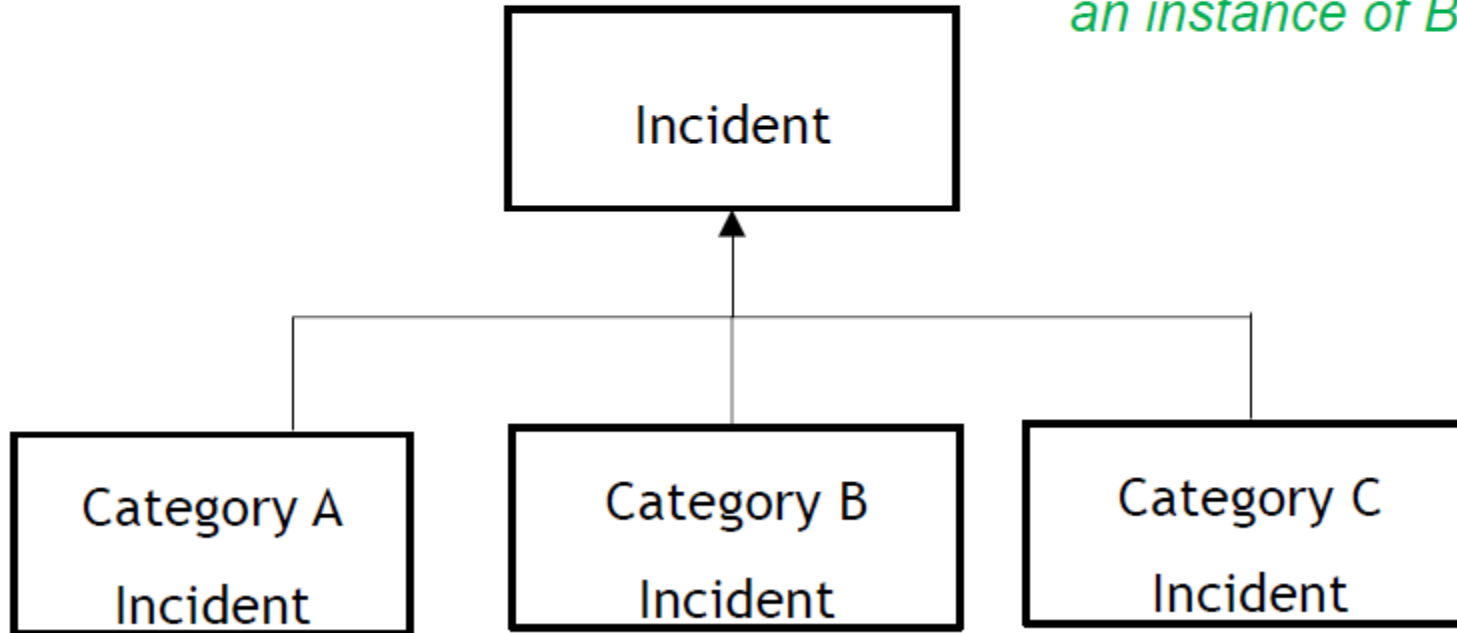
- Many confusions during software development are caused by imprecise and inconsistent terminology
 - people use different words for the same thing (synonyms), of the same word for different things (homonyms)
 - some words have no clear meaning or their meaning varies with context
 - people are often not aware of these inconsistencies
 - these situations lead to important misunderstandings
- A good conceptual model and glossary can reduce such confusion and facilitate communication throughout development
- A good conceptual model can also lead to software design that is easier to implement, understand, and change (cf. Domain-Driven Design)

Conceptual Model: Entity, Relationship, Attribute



Conceptual Modelling: Specialisation hierarchy

*Entity A specializes entity B
means every instance of A is
an instance of B*



Example glossary entries

Terms	Designation/Definitions
Entity Incident	Any event that may require an emergency ambulance response
location: Location	the incident' s location
Entity Ambulance	Any emergency vehicle managed by the ambulance service - this includes ambulance vans, cars, helicopters, motorbikes, bicycles.
Association Intervention [Ambulance, Incident]	An ambulance is in intervention at an incident when the ambulance is at the incident' s location
time: Time	the time at which the ambulance arrives at the incident' s location

Main criteria for good domain conceptual models

1. The model should be free of implementation concerns
2. The model should include all world phenomena used the description of goals, requirements and assumptions
3. The model should not include phenomena not used in the description of goals, requirements and assumptions

Example

Goal Achieve [Ambulance Intervention]

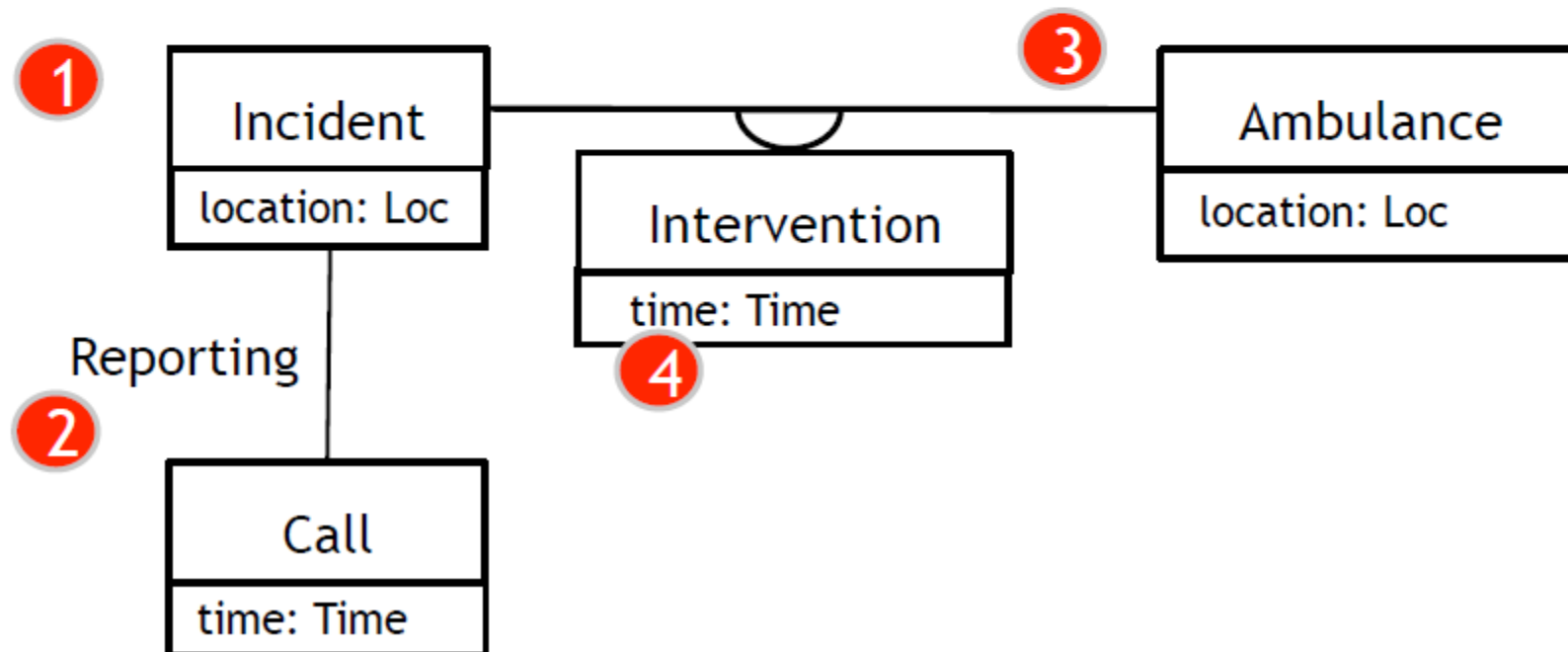
Desired Behaviour

GIVEN an incident has occurred **1**

WHEN the ambulance service receives a call reporting the incident **2**

THEN within 14 minutes **4**

a first ambulance arrives at the incident scene **3**



Exercise

Sketch a domain conceptual model for the following goal

Goal Achieve [Person at Destination Floor]

Definition

Given a person is at floor f

When the person wants to go to floor f'

Then eventually the person is at floor f'

Answer here after class

Exercise

Extend the domain conceptual model with the concepts used in these requirements

REQ Achieve [Lift request recorded]

Definition

Given the lift is not stopped at floor f

When a person at floor f presses the lift request button

Then the lift has a pending request for floor f

REQ Achieve [Lift starts when doors closed]

Definition

Given the lift is stopped at floor f

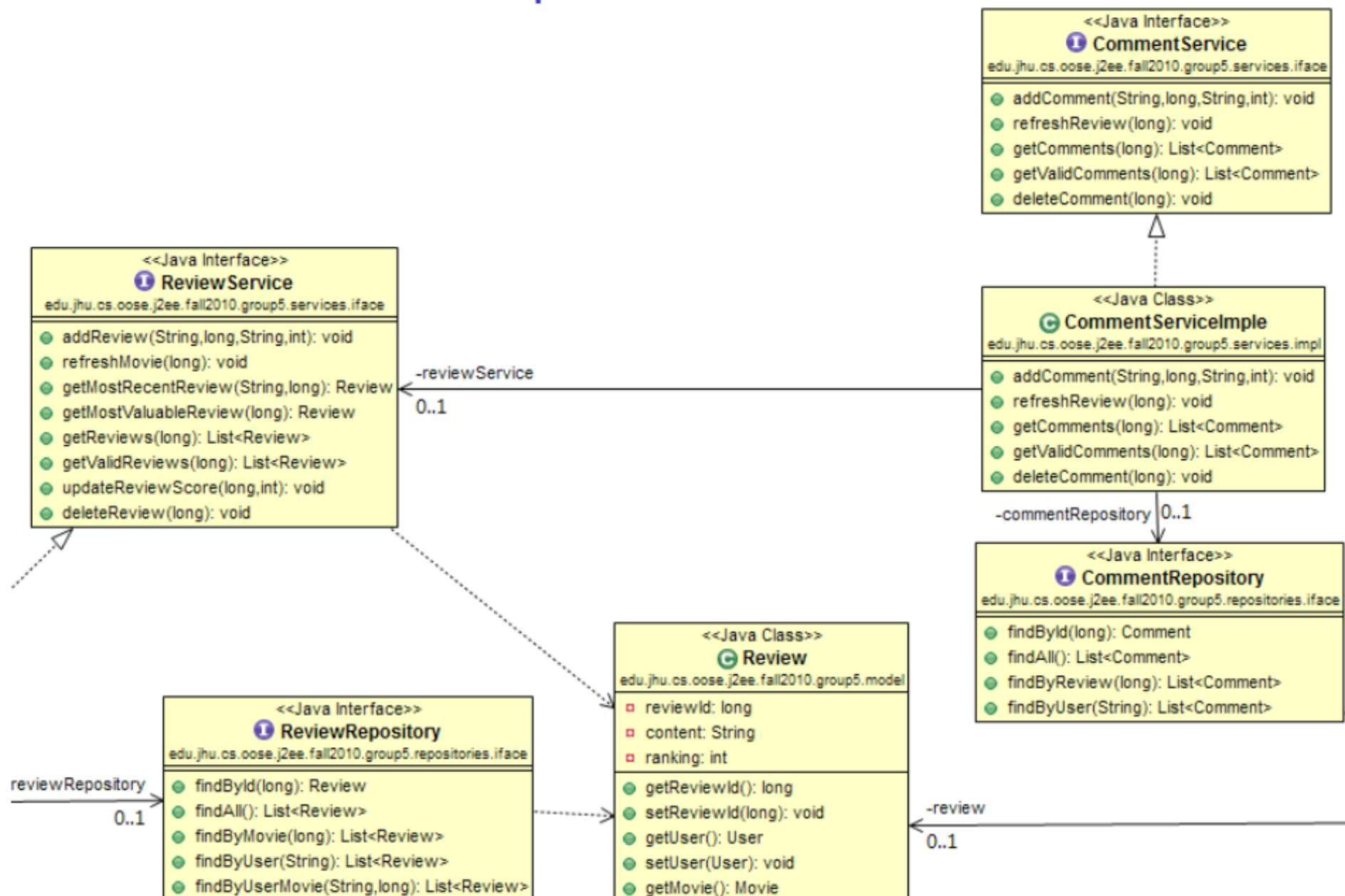
And the lift has a pending floor request

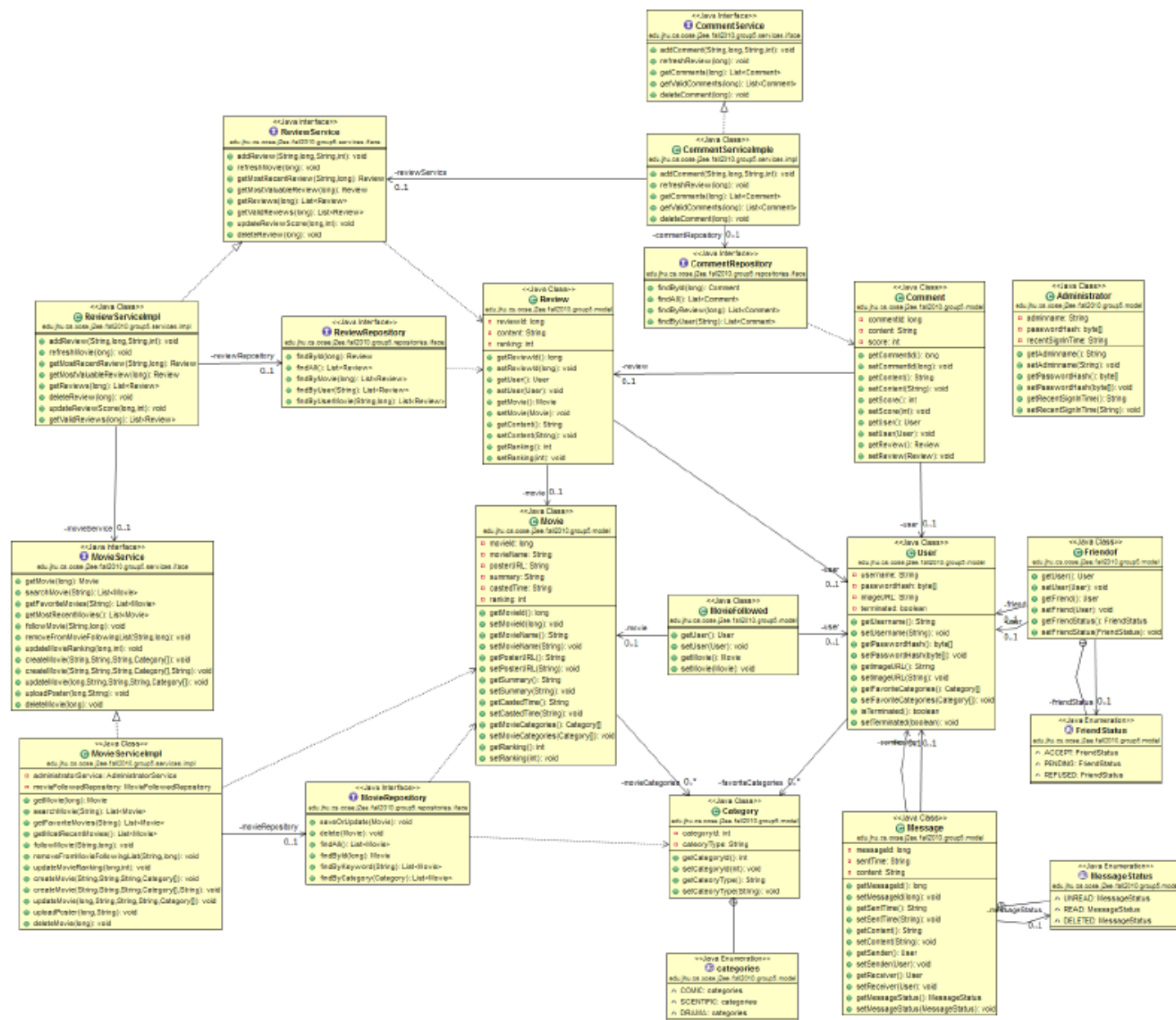
When the doors at floor f become closed

Then the lift starts moving in its direction of travel

DOs and DON'Ts of Conceptual Modelling

DON'T: include implementation concerns



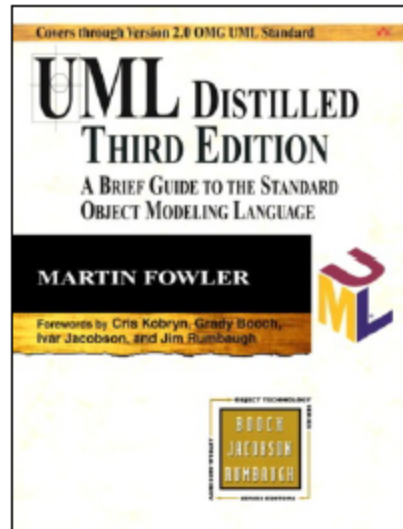


DO: Keep it simple and free of software elements

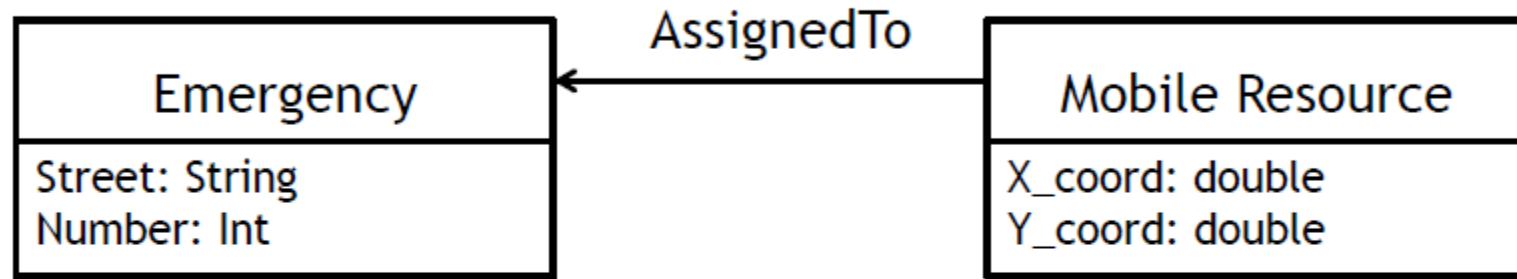


“I’ve found conceptual class diagrams very useful in exploring the language of a business. For this to work, you have to work hard on keeping software out of the discussion and keeping the notation very simple”

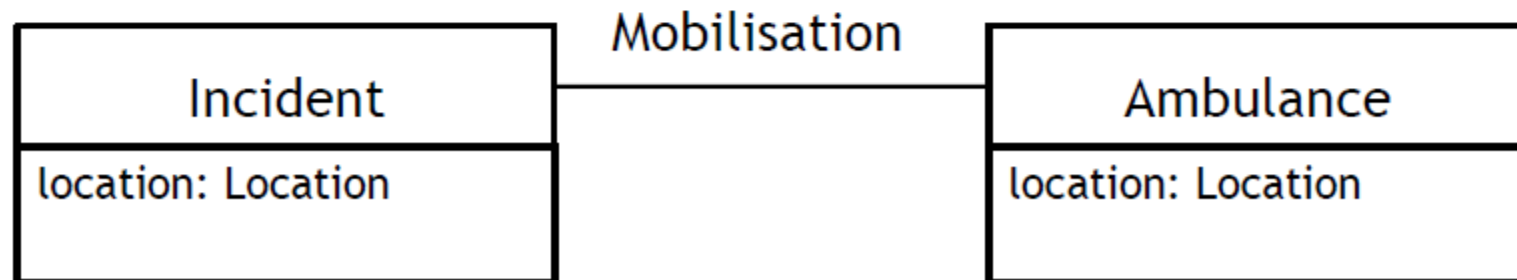
-- Martin Fowler in UML Distilled (3rd edition)



DON'T: use computer science jargon (or invent your own jargon)



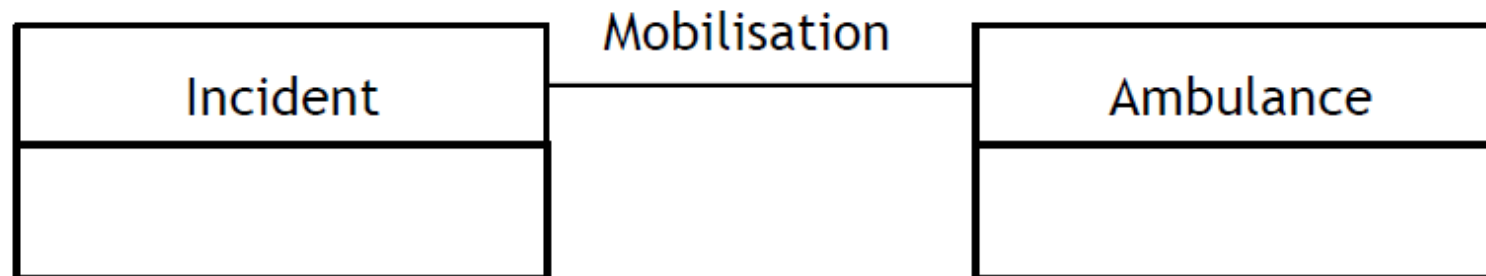
DO: use the words of your stakeholders



DON'T: use attributes as “pointer” to another entity

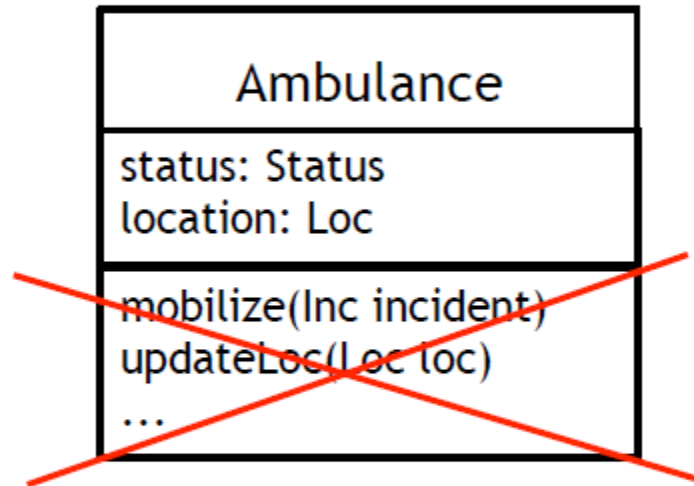


DO: use association instead of pointers



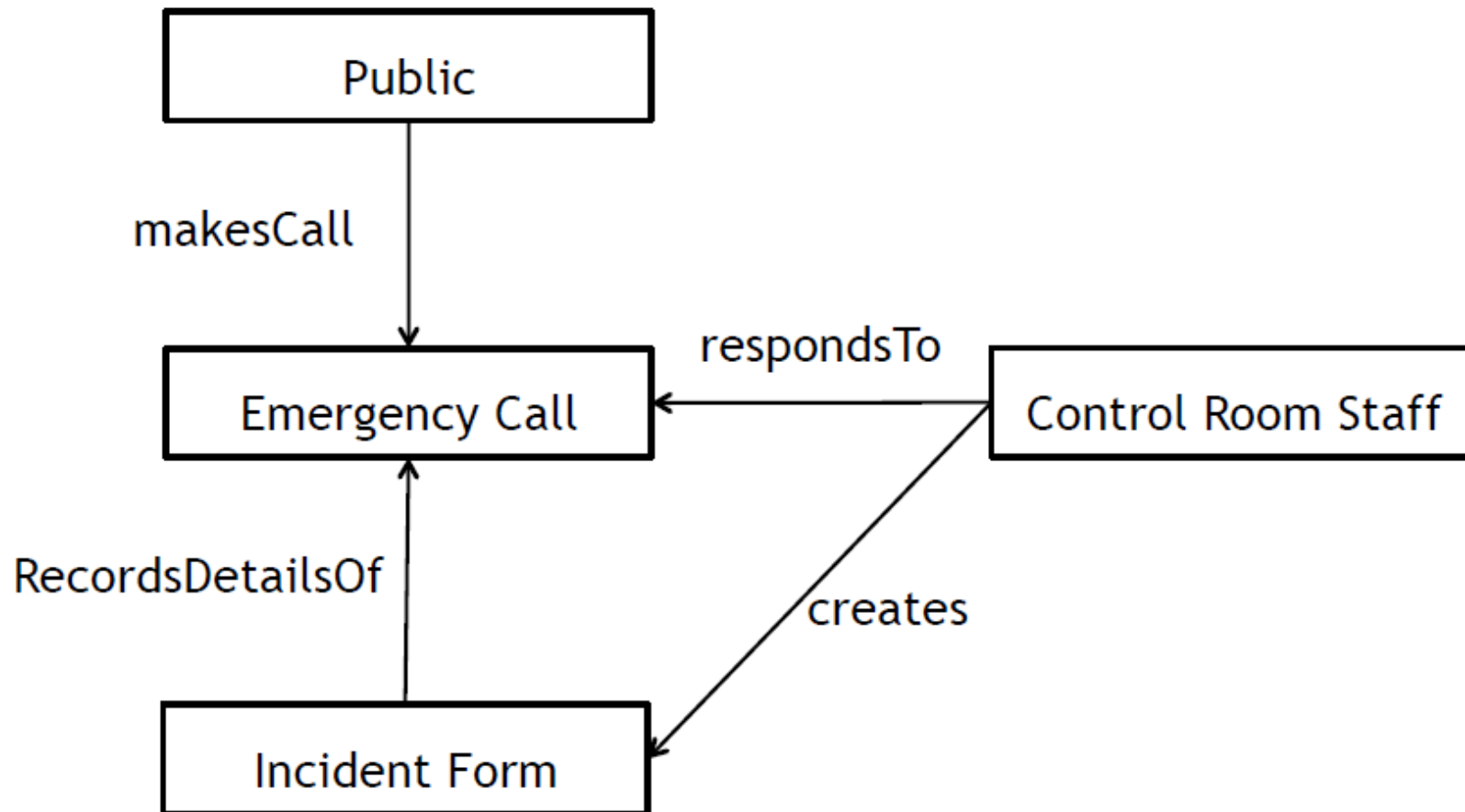
Question: how would you model the time at which an ambulance is mobilised to an incident?

DON'T: specify the methods of a class



Deciding what implementation class will perform what function is an implementation concern

DON'T: use associations to model activities

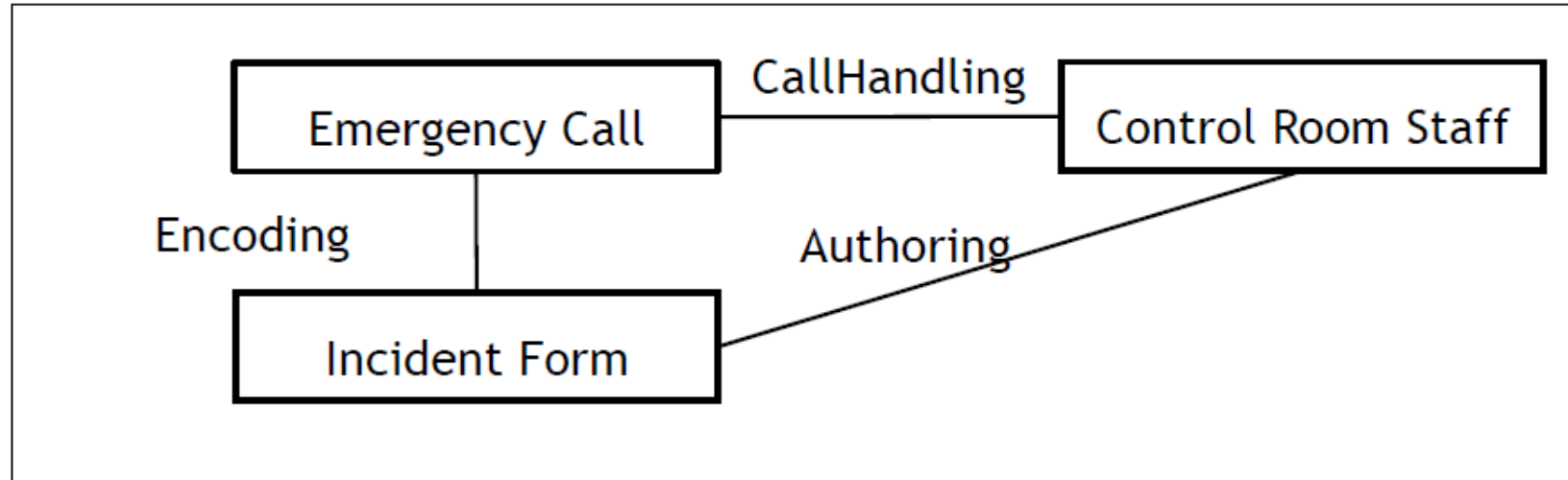


Just keep activities out of the conceptual model!

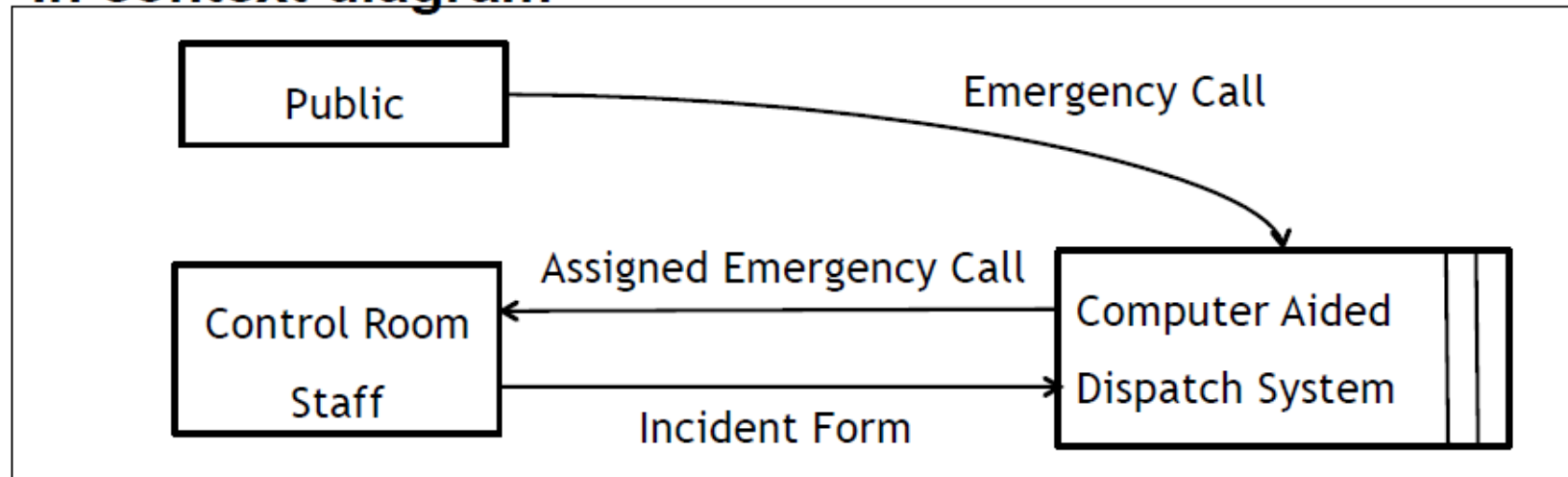
You can use other models to specify who does what (e.g. context diagrams, business processes, operation models)

DO: use context diagrams and contextual models as complementary views

In domain conceptual model

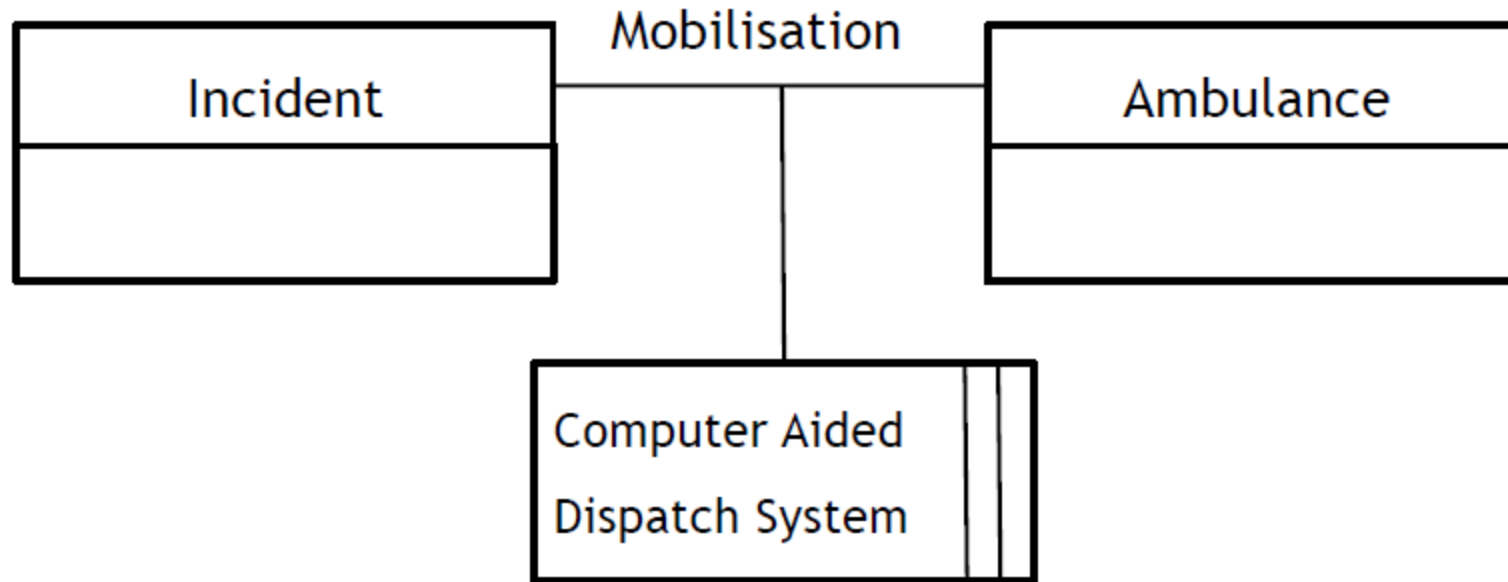


In context diagram



Frequently Asked Question

Can I include the Machine in my conceptual model?



Answer: Generally you should not (but there are exceptions)
You should instead use the context diagram to specify the machine's monitored and controlled phenomena
(as exercise, do this on the above example)

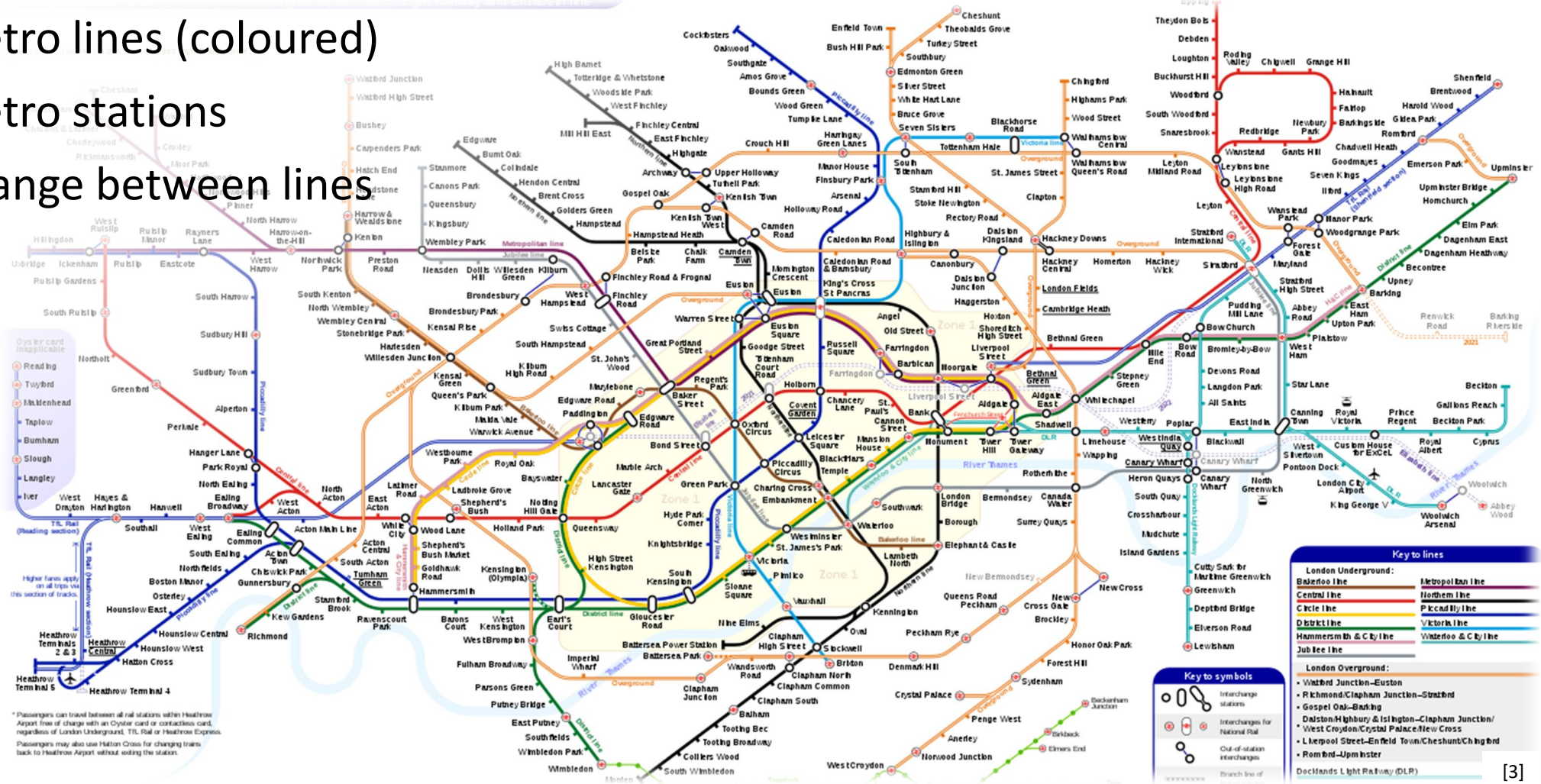
Purpose and Properties

- **Purpose**: A model is always **defined for a given purpose**
 - The **content** and **representation** of a model depends on this purpose
 - Different conceptual models support **different purposes**
 - The content documented in different conceptual models can **overlap**
- **Properties**:
 - **Reduction**: A conceptual model **documents only aspects** of the universe of discourse that are **relevant for the respective purpose** (and abstracts from non-relevant aspects).
 - **Extension**: A conceptual model can define **additional properties** which are **not observable in the universe of discourse**, e.g., to improve usability and understanding

Examples of Conceptual Models

Underground Map of London

- Metro lines (coloured)
- Metro stations
- Change between lines
- ...



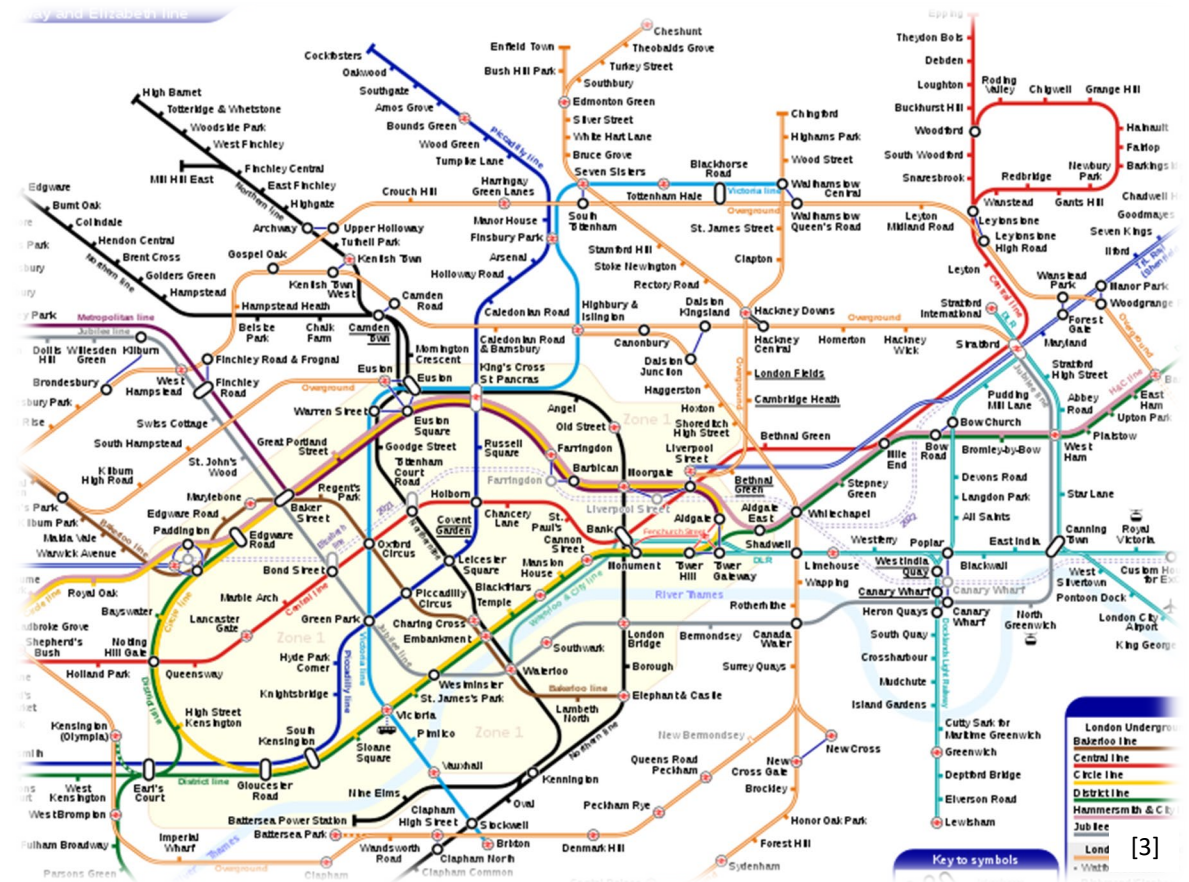
Examples of Conceptual Models

Different Abstractions of the Same Reality

Street map

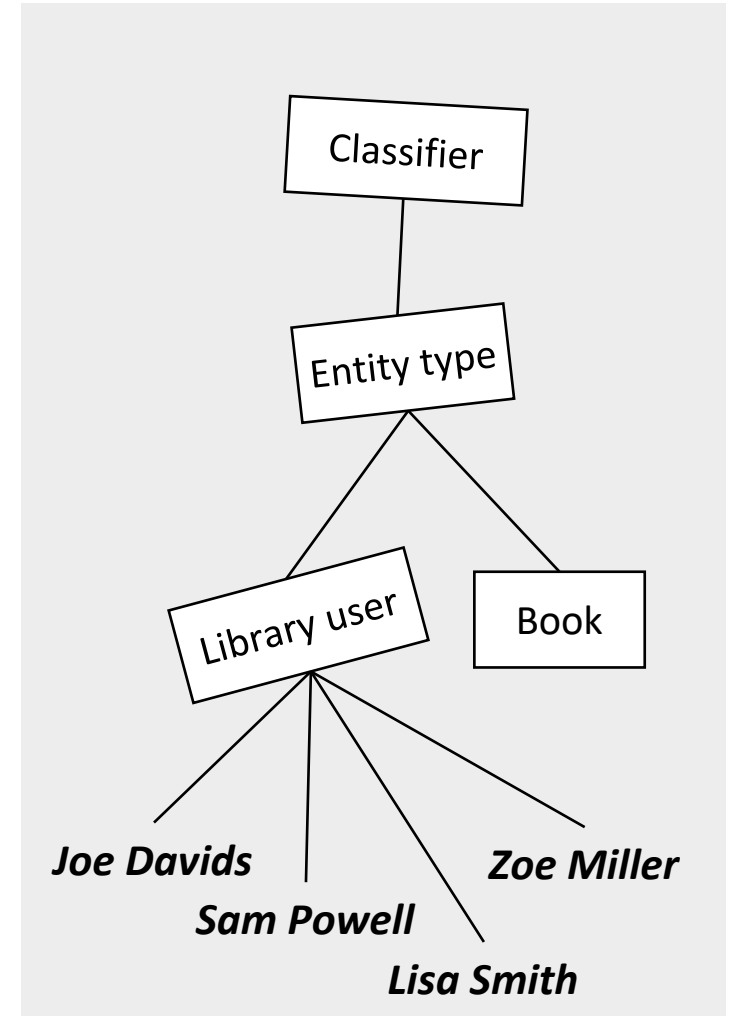
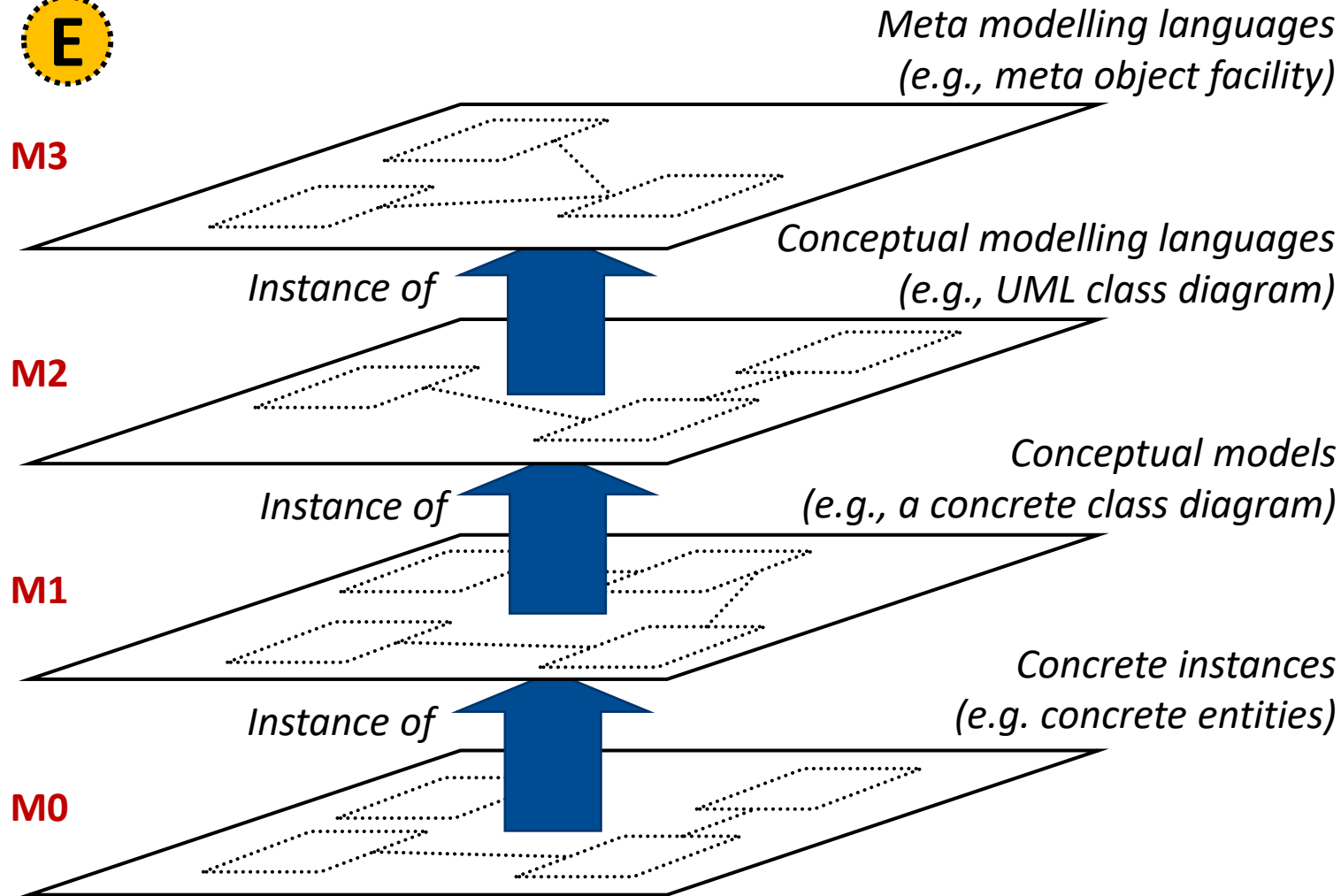


Underground map



Four Modelling Layers

Example: UML Class Diagrams



Focus of Each Modelling Layer M3 – M0

- M3: Meta modelling languages

Contains the definition of concepts which can be used for defining conceptual modelling languages

- M2: Conceptual modelling languages

Concrete conceptual model languages (with specific purposes) are defined at M2 using the meta modelling language constructs defined at M3

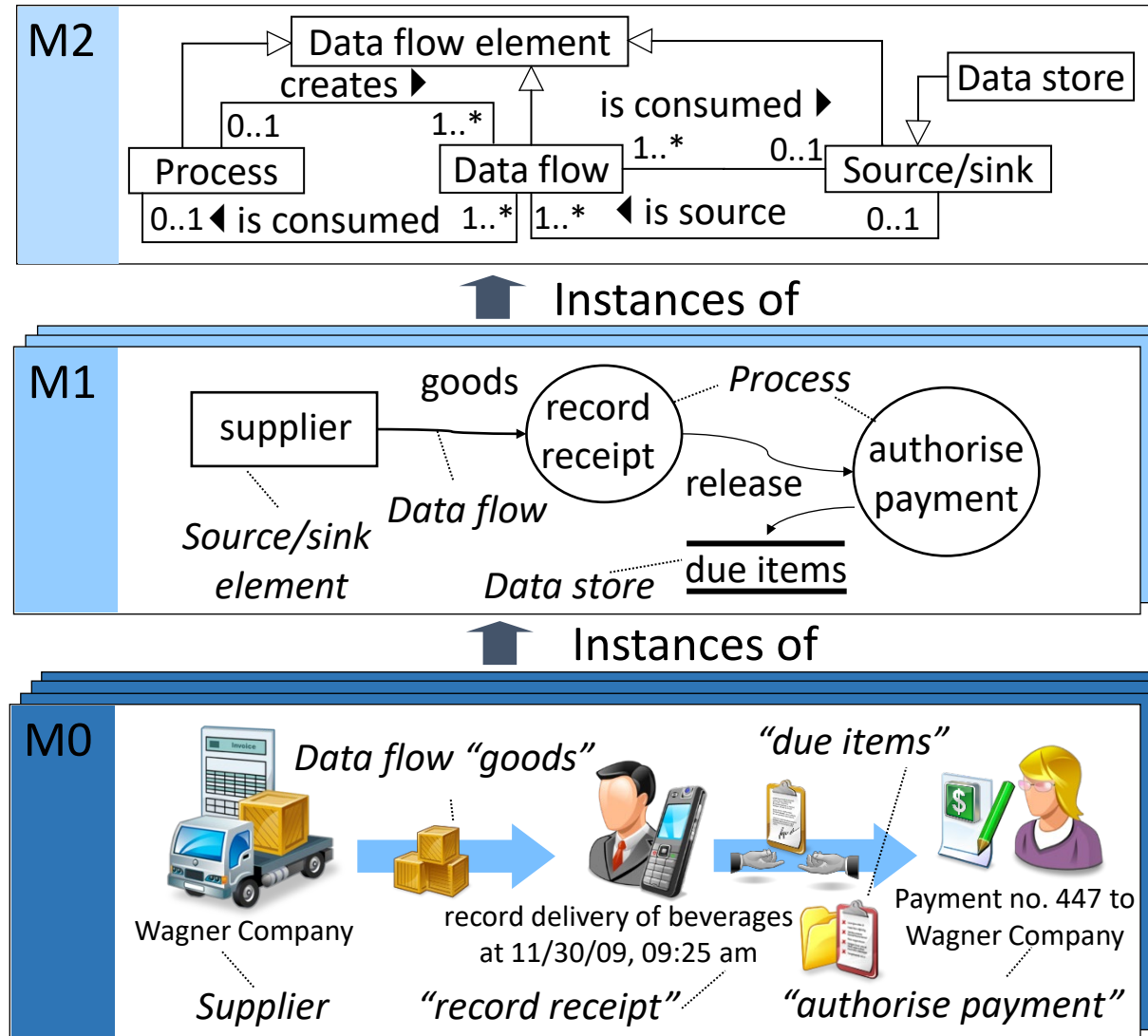
- M1: Concrete application-specific conceptual models

Application-specific conceptual models are defined using the conceptual modelling languages defined at M2

- M0: Concrete instances

Application-specific data is defined at the lowest level of the modelling hierarchy; e.g., recorded in a database, defined based on application specific conceptual models defined at M1

Example: Data Flow Diagrams



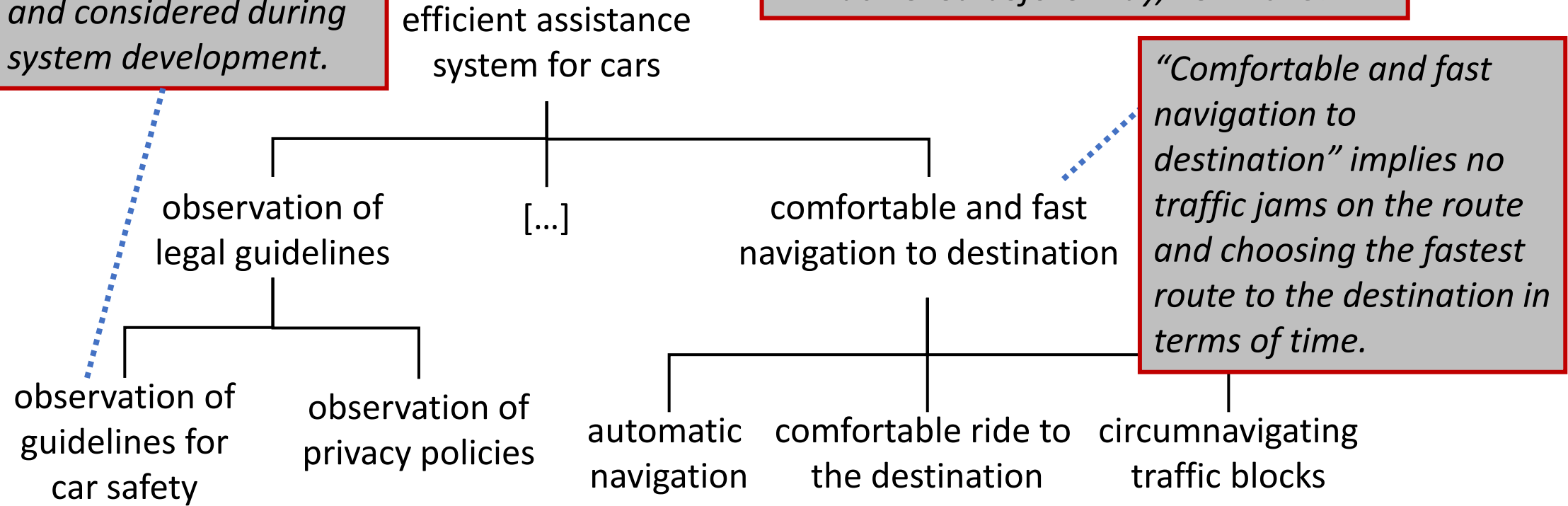
All icons from [1]

Textual Annotations of Model Elements

E

The ISO 26262 standard for functional safety should be continuously observed for changes and considered during system development.

Not all stakeholders have yet agreed on the goals. The agreement should be achieved before May, 13th 2023.



Interrelating Conceptual Models and Textual Artefacts

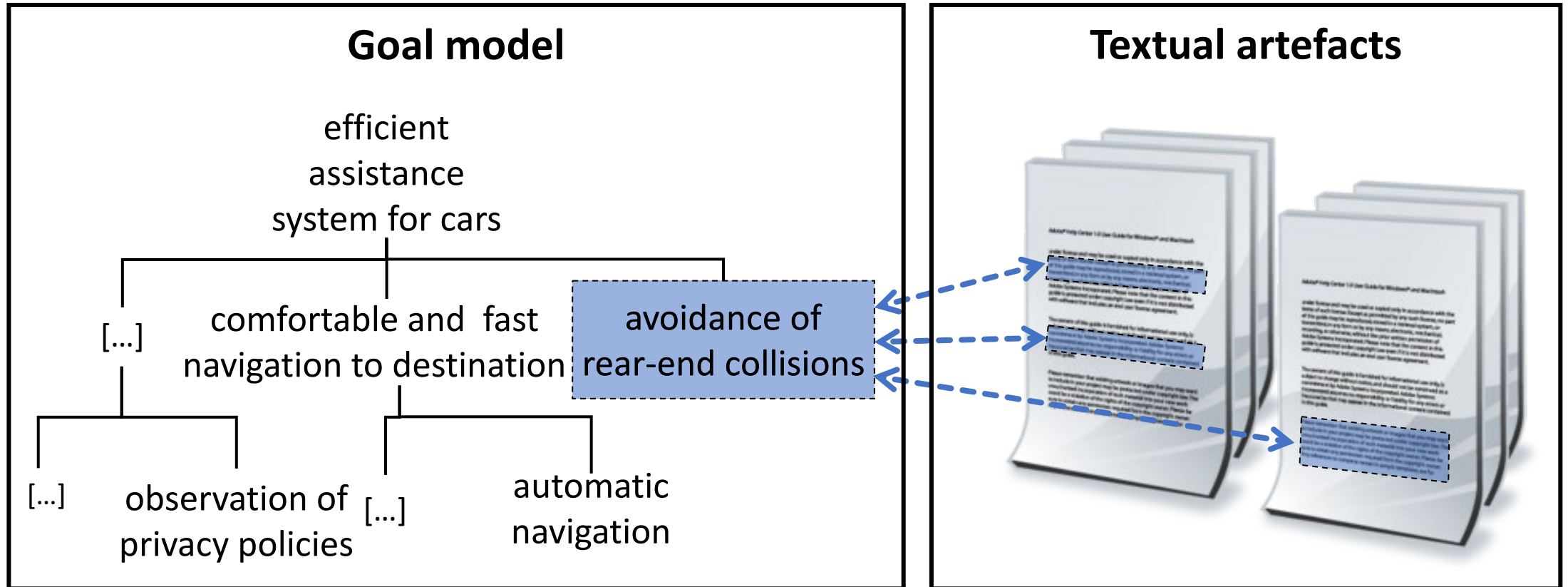
- Traceability relationships are used to link model elements with textual artefacts and vice versa
- Various types of traceability relationships can be used
 - Examples: An artefact
 - can be based on another artefact
 - can contain further details
 - can have a conflict with another artefact
- A traceability relationship can be used to link:
 - A single model element to one or multiple (parts of) textual artefacts
 - Multiple model elements to a single part of a textual artefact
 - Multiple model elements to multiple parts of textual artefacts

Details about traceability relationships
see lecture about Management

Combining Model-Based and Textual Documentations

Interrelation between a Model Element and a Set of Textual Artefacts

E

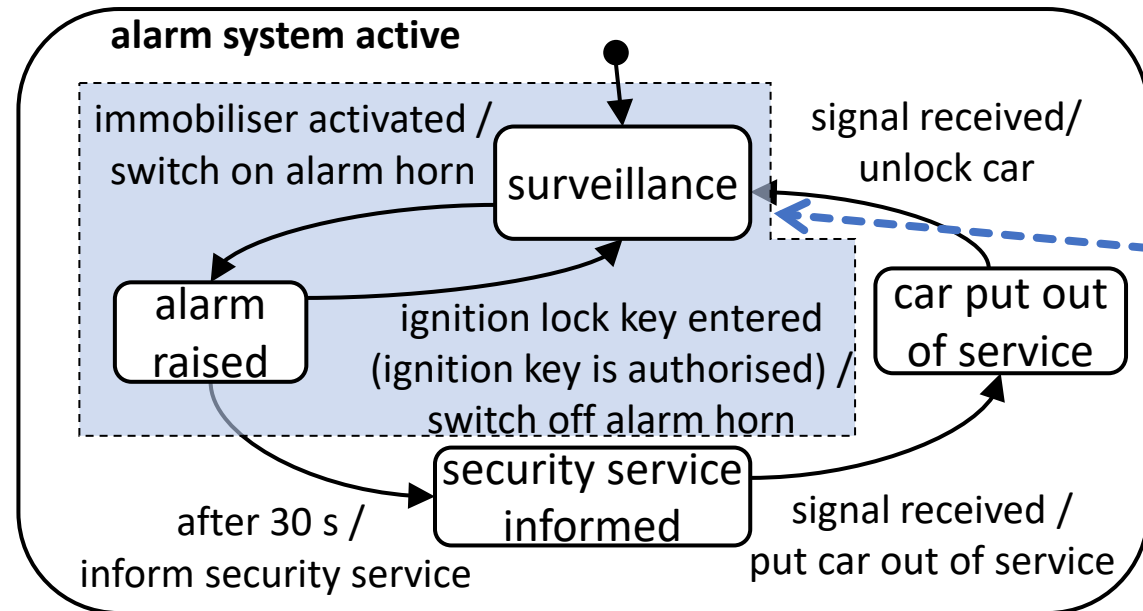


All icons from [1]

Interrelation between a Textual Artefact and a Set of Model Elements

E

Statechart



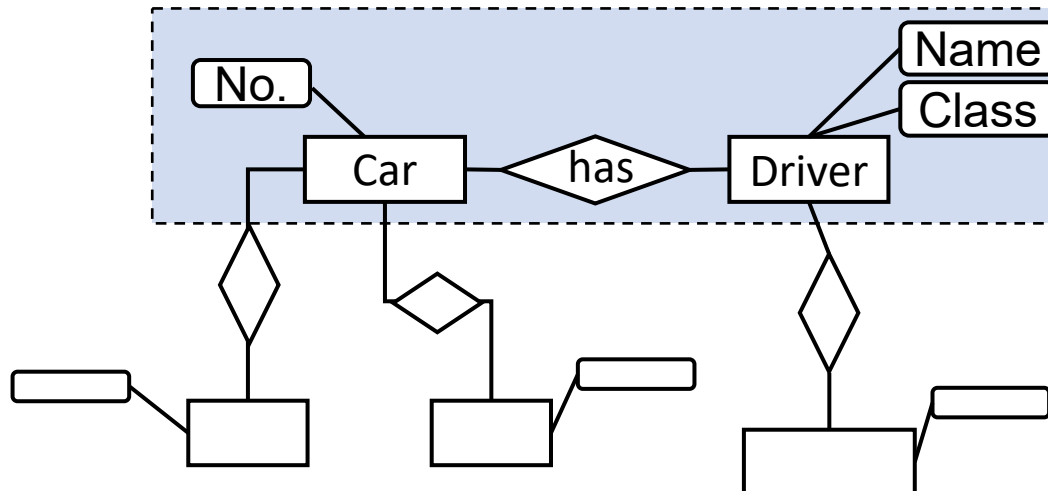
Textual artefacts



Interrelation between a Set of Textual Artefacts and a Set of Model Elements

E

Entity–Relationship Model

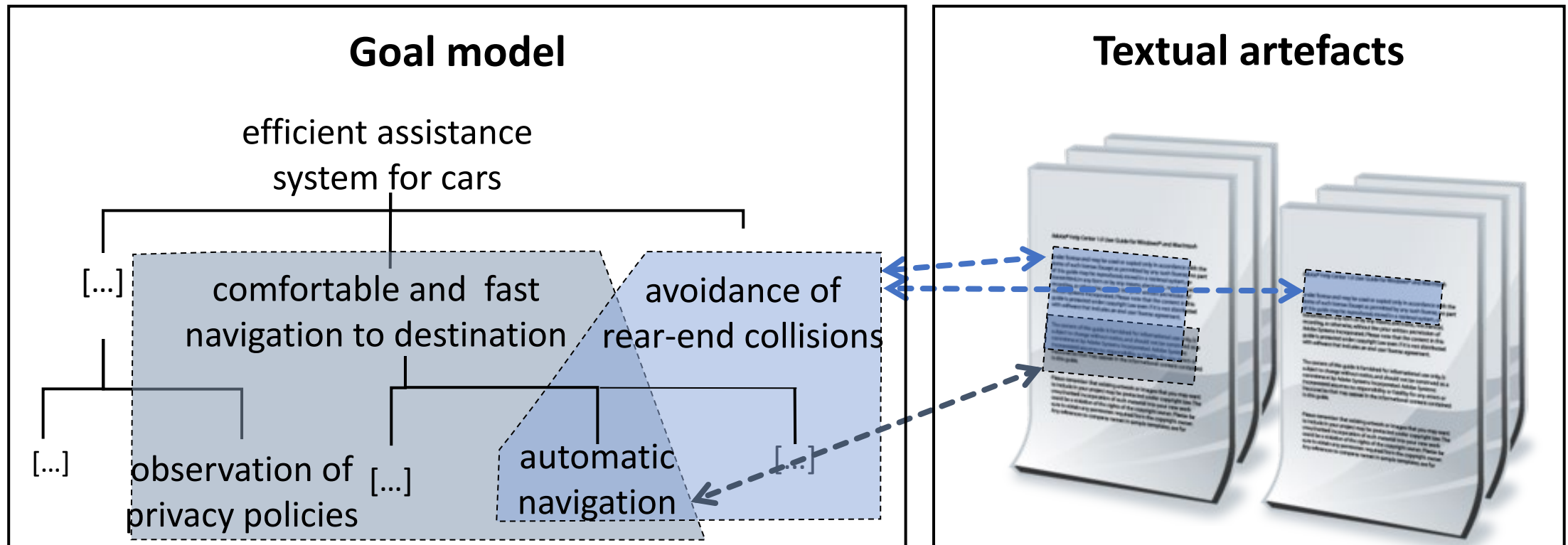


Textual artefacts



Sets can Overlap

E



Summary on Conceptual Models

- Are used in many areas of daily life
- Are created for a specific purpose and provide abstract (partial) symbolic representations of the current or a future universe of discourse
- Reduce the information documented about the universe of discourse
- Can add information not observable in the universe of discourse
- Are defined using a conceptual modelling language (i.e., the modelling constructs and rules defined in the language)
- Three kinds of model quality: syntactic, semantic and pragmatic quality

Summary (2)

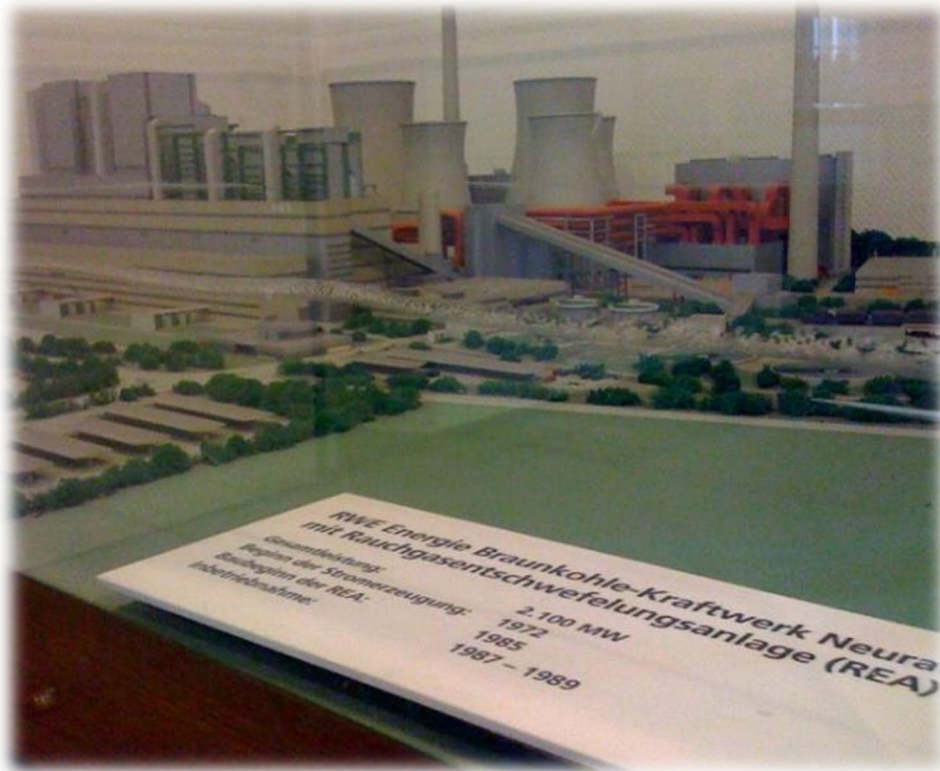
- Conceptual modelling languages
 - Define the abstract syntax, the concrete syntax and semantics of the modelling constructs
 - Are designed for a specific purpose
- Four modelling layers (meta-modeling) for defining meta languages, conceptual modeling languages, conceptual models and their instances
- Model-based and textual documentation can be combined, e.g., conceptual models can be annotated with textual representations, pictures and the like

Physical vs. Conceptual Models

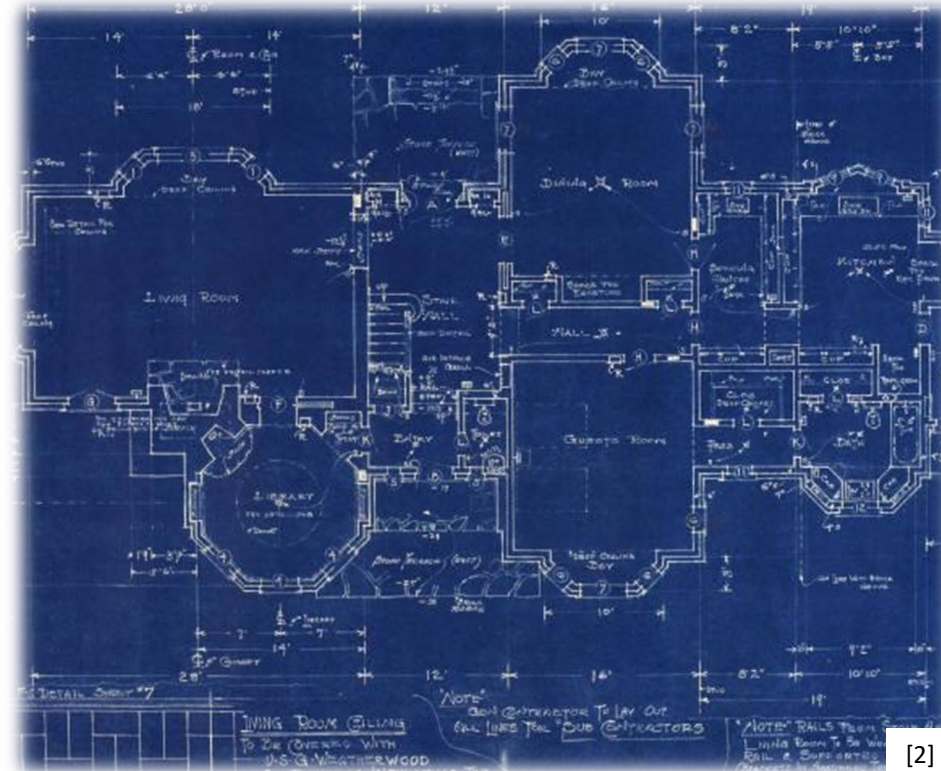
Examples



Physical Model



Conceptual Model



[2]

Literature References

- [Falkenberg et al. 1998] E. D. Falkenberg, W. Hesse, P. Lindgreen, B. E. Nilsson, J. L. Han Oei, C. Rolland, R. K. Stamper, F. J. M. Van Assche, A. A. Verrijn-Stuart, K. Voss: A Framework of Information System Concepts – The FRISCO Report. IFIP Report, 1998.
- [Ogden and Richards 1923] C. K. Ogden, I. A. Richards: The Meaning of Meaning – A Study of the Influence of Language upon Thought and of the Science of Symbolism. Routledge & Kegan, London, 1923 (new edition available: Harcourt Brace Jovanovich, 1989).
- [Seidewitz 2003] E. Seidewitz: What Models Mean. IEEE Software, Vol. 20, No. 5, IEEE Press, Los Alamitos, 2003, pp. 26-32.
- [Stachowiak 1973] H. Stachowiak: Allgemeine Modelltheorie. Springer, Wien, 1973.
- [Heine 1824 / Silcher 1837] H. Heine: Die Loreley – Text, 1824. F. Silcher: Liedfassung, 1837.

Image References

- [1] Licensed by <http://www.icons shock.com/>
- [2] Provided by Microsoft Office
- [3] London Tube Map, by Sameboat (Own work) [CC BY-SA 4.0 (<http://creativecommons.org/licenses/by-sa/4.0>)], via Wikimedia Commons. https://commons.wikimedia.org/wiki/File%3ALondon_Underground_Overground_DLR_Crossrail_map.svg, accessed on November 4, 2021.
- [4] Satellite view of London, bird's-eye view, [Public domain], via Wikimedia Commons. <https://commons.wikimedia.org/wiki/File%3APIA04301.jpg>, accessed on March 5, 2015.
- [5] Inside Guide to London.
- [6] Jans House model, by Dusan Jurkovic (Own work) [CC BY-SA 4.0 (<http://creativecommons.org/licenses/by-sa/4.0>)], via Wikimedia Commons. http://upload.wikimedia.org/wikipedia/commons/2/2b/Dusan_Jurkovic_Januv_dum.jpg, accessed on March 5, 2015.
- [7] Belton Plan, by Giano at en.wikipedia [Public domain], from Wikimedia Commons. http://upload.wikimedia.org/wikipedia/commons/f/f7/Belton_Plan.jpg, accessed on March 5, 2015.
- [8] From Adobe Stock, https://stock.adobe.com/de/images/kreisverkehr-es-geht-rund/378047691?prev_url=detail

Legend

 Definition

 Example

Incorrect